

3.1 Biological molecules

All life on Earth shares a common chemistry. This provides indirect evidence for evolution.

Despite their great variety, the cells of all living organisms contain only a few groups of carbon-based compounds that interact in similar ways.

Carbohydrates are commonly used by cells as respiratory substrates. They also form structural components in plasma membranes and cell walls.

Lipids have many uses, including the bilayer of plasma membranes, certain hormones and as respiratory substrates.

Proteins form many cell structures. They are also important as enzymes, chemical messengers and components of the blood.

Nucleic acids carry the genetic code for the production of proteins. The genetic code is common to viruses and to all living organisms, providing evidence for evolution.

The most common component of cells is water; hence our search for life elsewhere in the universe involves a search for liquid water.